FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2021

TEST DATE: 07-03-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. $B = -\frac{Vdp}{dV}$

$B = \frac{1}{\alpha}$

2. C

Sol. Let the particle strikes the inclined plane at point P at an angle θ from the inclined plane

$v\sin\theta = v_0\sin 30°$ …(i)

$v\cos\theta = v_0\cos 30° - g\sin 30°\left(\frac{2v_0\sin 30°}{g\cos 30°}\right)$ …(ii)

Solving equation (i) and (ii)

$\theta = 60°$ and $v = \frac{10}{\sqrt{3}} ms^{-1}$

3. A

Sol. for $v > \frac{\mu_0}{\lambda}, \mu = 0$

$a = \frac{F}{m}.$

4. B

Sol. U + KE = Const.

U = constant (if speed is constant)

$y^2 = 2kx$. Parabola

5. B

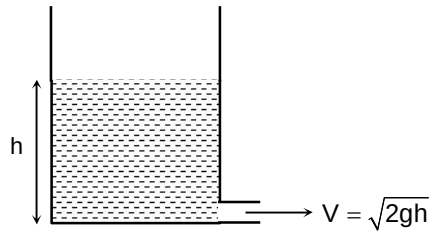
Sol. C.M. will lie on y axis because it is axis of symmetry.

$$Y_{cm} = \frac{\int ydm}{\int dm}$$

6. A

Sol. $\rho hA.\frac{dv}{dt} = \rho a.2gh$

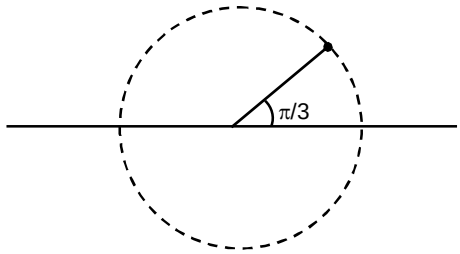
$\Rightarrow \quad v = \frac{2a}{A}.gt$



7. C

Sol. $\frac{5\pi}{6} = \omega t$

$t = \frac{5\pi}{6\omega}$



8. C

Sol. $f = \frac{f_0(c+v)}{c} \Rightarrow f' = \frac{f}{2}\left(\frac{2c-v}{c+v}\right)$

9. B

Sol. $L = mr^2.\omega$

$= mr^2.\frac{2\pi}{T} \Rightarrow \frac{L_A}{L_B} = \frac{m_A}{m_B}\frac{r_A^2}{r_B^2}\left(\frac{r_B}{r_A}\right)^{3/2}$

$= \frac{1}{2} \times \frac{16}{1} \times \frac{1}{8}$

10. B

Sol. $E_{net} = \sqrt{E_x^2 + E_y^2}$

11. D

Sol. $R_{eg} = \frac{100}{15}\Omega$

Current through battery $i = \frac{50 \times 15}{100}$

i = 7.5 A.

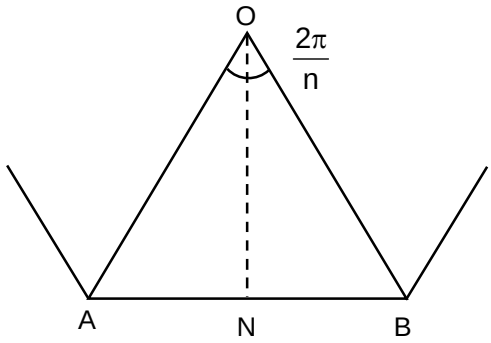
12. C

Sol. $AB = \frac{\ell}{n}$

$ON = \frac{\ell}{2n}\cot\frac{\pi}{n}$

$B_n = \frac{n.\mu_0 i}{4\pi\frac{\ell}{2n}\cot\left(\frac{\pi}{n}\right)}.2\sin\frac{\pi}{n}$

$= \frac{\mu_0 n^2 i}{\pi\ell}\frac{\sin^2\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right)}$



13. D

Sol. $\frac{V_1}{R} > \frac{V_2}{R}$

$C_1V_1 > C_2V_2$

14. A

Sol. $\frac{1}{2}Li_0^2 + \frac{1}{2}.\frac{3i_0^2}{C}.LC = \frac{1}{2}Li_{max}^2$

$I_{max} = 2i_0$

15. C

Sol. $E_0 = B_0C$ and $\left(\vec{E}\times\vec{B}\right)$ gives the direction of propagation of wave.

16. D

Sol. $E_g = h\nu$

$\lambda = \frac{1240}{1.5}nm$

17. A

Sol. $\rho = \frac{M}{\frac{4\pi r^3}{3}} \Rightarrow \frac{\Delta\rho}{\rho} = \frac{3\Delta r}{r}$

18. C

Sol. $\Delta\theta = \frac{2\lambda}{d}$

$= \frac{2\times400\times10^{-9}}{40\times10^{-6}}$

$= 2 \times 10^{-2} = 0.02$ rad.

19. A

Sol. $\vec{F} = q\vec{E} + q\left(\vec{V}\times\vec{B}\right)$

$$\vec{a} = \frac{qE_0\hat{j}}{m} - \frac{qV_yB_0\hat{k}}{m}$$

20. B

Sol. $\lambda_{min} = \frac{hc}{eV}$

$\frac{1242}{15000}$nm

= 0.082 nm.

SECTION – B

21. 00002.00

Sol. $\frac{Q}{T}$ = Constant (For Carnot cycle).

22. 00003.00

Sol. $K_1 = \frac{1}{2}I_{cm}\omega^2 = \frac{mv^2}{4}$

$K_2 = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2 = \frac{3mv^2}{4}$

23. 00050.00

Sol. $\cos\phi = \frac{R}{\sqrt{X_C^2 + R^2}} \Rightarrow X_C = 200\,\Omega$

$C = \frac{1}{200 \times 100}$

= 50 μF

24. 00140.00

Sol. Q = (120 × 6) + (80 × 6.5) – (200 × 5.5)

Q = 720 + 520 – 1100 = 140 MeV

25. 00010.00

Sol. Image formed by the convex lens should be at the centre of curvature of the convex mirror.

26. 00020.00

Sol. $i = \frac{\varepsilon}{x_C}$

27. 00012.50

Sol. $\mu g = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{\mu g}$

28. 00001.50

Sol. $P = 2P_l + P_m$

29. 00001.75

Sol. $A_r = \left(\frac{v_2 - v_1}{v_2 + v_1}\right) A_i = \left(\frac{1 - \sqrt{\frac{\mu_2}{\mu_1}}}{1 + \sqrt{\frac{\mu_2}{\mu_1}}}\right) A_i$

$A_r = \left(\frac{1 - \frac{1}{3}}{1 + \frac{1}{3}}\right) A_i = \frac{A_i}{2} = 1.75 \text{ cm}$

$A_r = 1.75 \text{ cm}$

30. 00007.00

Sol. $v = 4 + 6t - 3t^2$

When v is maximum

$a = 0 \Rightarrow a = 6 - 6t$

$t = 1 \text{ sec}$

$v_{max} = 7 \text{ m/s}$

Chemistry

PART – B

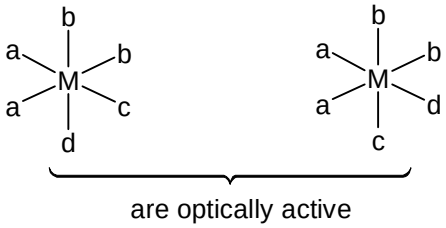
SECTION – A

31. B

Sol. Cerussite is a mineral consisting of lead carbonate and is an important ore of lead.

32. C

Sol.



33. B

34. C

Sol. The complex $Fe_4\left[Fe(CN)_6\right]_3.xH_2O$ is Prussian blue in colour.

35. B

Sol. The compound B will show geometrical isomerism along ring.

36. C

Sol. Ionisation energy depends on electronic configuration and effective nuclear charge

$Be^{2+} = 1s^2$ $C^{2+} = 1s^2 2s^2$ $N^{2+} = 1s^2 2s^2 2p^1$ $B^{2+} = 1s^2 2s^1$

37. B

Sol. $3CH_3COOH + PCl_3 \longrightarrow 3CH_3COCl + H_3PO_3$

Moles of $CH_3COOH = \frac{3.6}{60} = 0.06$

So, 0.02 moles of H_3PO_3 is produced

Equivalents of H_3PO_3 = Equivalents of base

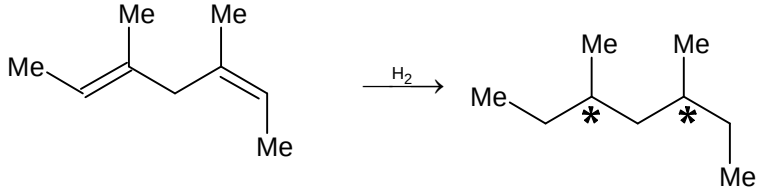
$0.02 \times 2 = \frac{200}{1000} \times 0.1 \times 2$

38. C

Sol. $CH_3 - \overset{O}{\overset{\|}{C}} - NH_2 \xrightarrow{P_2O_5} CH_3 - C \equiv N \xrightarrow[(2)\,H_3O^+]{(1)\,CH_3MgBr} CH_3 - \overset{O}{\overset{\|}{C}} - CH_3 \xrightarrow[NaOH]{I_2} CHI_3 + CH_3 - \overset{O}{\overset{\|}{C}} - O^-$

39. C

Sol.



Compound is symmetrical so the total number of stereoisomers is 3.

40. B

Sol. For MnO_4^{3-} its standard reduction potential is greater value than standard oxidation potential hence it will undergo disproportionation reaction due to overall standard potential is positive.

$MnO_4^{3-} + e^- \longrightarrow MnO_2 \quad E^o = 0.93$

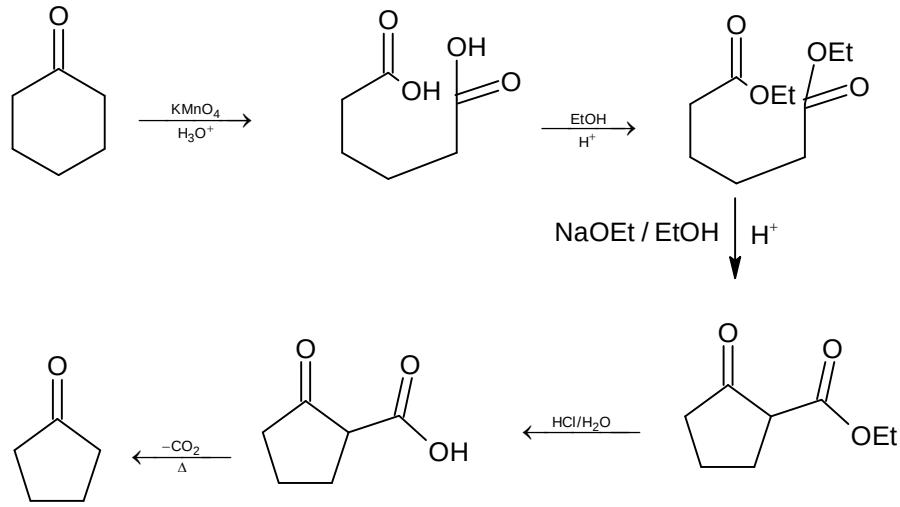
$MnO_4^{3-} \longrightarrow MnO_4^{2-} + e^- \; E^o = -0.27$

$2MnO_4^{3-} \longrightarrow MnO_2 + MnO_4^{2-}$

$E^o_{total} = 0.93 - 0.27 = 0.66\ V$

41. C

Sol.



42. A

Sol.

$A(g) \rightleftharpoons B(g) + C(g)$

$t = 0$	a		
at equilibrium	$a - 0.5a$	$0.5a$	$0.5a$

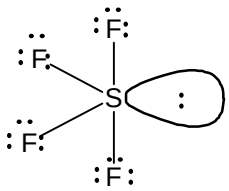
$K_P = \frac{P_B P_C}{P_A} = \frac{1}{3} = 0.33$

$\Delta G = -RT \ln K_P$

= 4.61 kJ/mol

43. D

Sol.



44. D

Sol.

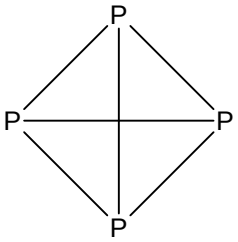
$$C_6H_5 - C \equiv CH \longrightarrow C_6H_5 - \overset{O}{\overset{\|}{C}} - CH_3 \xrightarrow[(2)\ H^+, \Delta]{(1)\ PhCHO, OH^-} C_6H_5 - \overset{O}{\overset{\|}{C}} - CH = CH - Ph$$

$$\xrightarrow[\Delta]{NH_2-NH_2/OH^-} C_6H_5CH_2CH = CHPh \xrightarrow[(2)\ Zn/H_2O]{(1)\ O_3} C_6H_5CH_2CHO$$

45. B

46. A

Sol.



Atomicity = 4
Total bonds = 6

47. B

Sol. $V = \frac{h}{m\lambda} = \frac{6.6\times10^{-34}}{9.1\times10^{-31}\times13.4\times10^{-10}}$

$V = 2.18\times10^{6}\frac{Z}{n}\,m/s$

48. C

Sol. Aspartic acid $\rightarrow Asp\left(-CH_2-COOH\right)$

Phenylalanine $\rightarrow Phe\left(-CH_2-C_6H_5\right)$

Methionine $\rightarrow Met\left(-CH_2-CH_2-S-CH_3\right)$

49. B

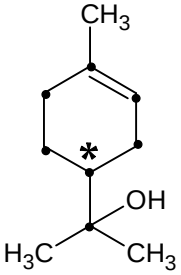
Sol. Li_2O and MgO do not combine with excess oxygen to give superoxide.

50. C

SECTION – B

51. 00000.10

Sol.



52. 00001.95

Sol. $ppm = \frac{\text{Mass of Ca in mg}}{\text{Volume of solution in litres}}$

$9.50 = \frac{\text{mass of Ca in mg}}{50\times10^{-3}}$

Mass of Ca in mg = 0.475 mg

Moles of Ca $= \frac{0.475\times10^{-3}}{40}$

Moles of Ca = Moles of $Ca(NO_3)_2$

Mass of $Ca\left(NO_3\right)_2 = \frac{164\times0.475\times10^{-3}}{40} = 1.95\times10^{-3}$

= 1.95 mg

53. 00033.33

Sol. $PV = \frac{1}{3}mNu_{rms}^2$

$P = \frac{1\times10^{-25}\times10^{23}}{3\quad\quad10^{-3}}\times\left(10^3\right)^2$

$P = 33.33\times10^5$ Pa.

54. 00014.90

Sol. $\Delta S_{system} = -nR\ell n\frac{P_2}{P_1}$

$= -1\times8.314\ell n\frac{30}{5}$

= - 14.90

$\Delta S_{surrounding} = -\Delta S_{system} = 14.90$

55. 00000.37

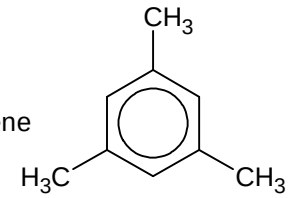
Sol. $\Delta T_f = \left(i_{NaCl}\ m_{NaCl} + i_{sucrose}\ m_{sucrose}\right)K_f$

$\Delta T_f = \left(2\times0.05 + 1\times0.1\right)1.86$

$\Delta T_f = 0.37$

56. 00090.00

Sol.

P is Mesitylene 

Molecular weight of C_9H_{12} =

$\%C = \frac{12\times9\times100}{120} = 90$

57. 67258.17

Sol. $T_1 = 310$ K $\quad T_2 = 340$ K

$K_{T_2} = 10K_{T_1}$

$\ell n\left(\frac{K_{T_2}}{K_{T_1}}\right) = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$\ell n10 = \frac{E_a}{8.314}\left(\frac{1}{310} - \frac{1}{340}\right)$

$E_a = 67258.17\,J/mol$

58. 00139.43

Sol. $d = \frac{Z\times M}{N_A\times a^3}$

Putting all values, Z = 2, then structure is bcc.

$\sqrt{3}a = 4r$

$r = 139.43\times10^{-12}$ m

59. 00008.78

Sol. m mole of ammonia in 20 ml = 2
m mole of NaOH in 20 ml = 1
m mole of HCl = 2.5

$$NH_3 + HCl(\text{remaining}) \longrightarrow NH_4Cl$$

2	$\frac{3}{2}$	0
0.5	0	$\frac{3}{2}$

$$pOH = pK_b + \log\frac{3/2}{0.5}$$

pOH = 5.22
pH = 8.78

60. 00001.50

Sol. PCl_5 in solid state exist as $[PCl_4]^+ [PCl_6]^-$

Mathematics

PART – C

SECTION – A

61. C

Sol. Let $g(x)=\frac{x^3}{(1-x)^3+x^3} \Rightarrow g(x)+g(1-x)=1$

$\Rightarrow 2f(101)=\sum_{r=0}^{101}\left(f(x_r)+f(x_{101-r})\right)=102 \Rightarrow f(101)=51$

62. B

Sol. $2P(r)=\int_0^{\pi/2} dx \Rightarrow P(r)=\frac{\pi}{4} \ \forall\ r \in R$

63. D

Sol. $P(E)=\frac{\frac{1}{2}\times\frac{1}{3}}{\frac{1}{2}\times\frac{1}{3}+\frac{1}{4}\times\frac{1}{6}+\frac{1}{8}\times\frac{1}{2}}=\frac{\frac{1}{6}}{\frac{13}{48}}=\frac{8}{13}$

64. C

Sol. $P(k) = P(k+1) \Rightarrow h = k^2 + 5k + 2$

$\Rightarrow P(k) = 2(k^2 + k - 2)$

$k = 1, -2 \Rightarrow k = -2$ as $k \neq 1$

$\Rightarrow h = -4$

65. D

Sol. $2|c| \geq |a+b+c| + |a-b+c| \geq 2|a+c|$

$\Rightarrow |a+c| \leq 2$ as $|c| \leq 1$

66. C

Sol. $2(a - \cos b) = a^2 + 2a - 1 = (a+1)^2 - 2$

$-1 \leq a - \cos b \leq \left(\sqrt{3}+1\right)$ as $0 \leq a^2 \leq 3$

67. C

Sol. Number of ways of selecting three collinear points = $10 \cdot {}^4C_3 + 4 \cdot {}^3C_3 = 44$

68. B

Sol. $5 - 3|x-2|^2 - 2|x-2| \geq 0 \Rightarrow |x-2| \leq 1 \Rightarrow x \in [1, 3]$

69. D

Sol. $2 \leq |z| \leq 5$ and $|z+3| \leq |z-3|$

70. B

Sol. $\lim_{x\to\infty} e^{-\sec\left(\tan^{-1}x\right)} \Rightarrow \lim_{h\to\frac{\pi^-}{2}} e^{-\sec h} = 0$

$\lim_{x\to-\infty} e^{\tan\left(\sec^{-1}x\right)} \Rightarrow \lim_{h\to\frac{\pi^+}{2}} e^{\tan(h)} = 0$

A = 0 and B = 0

71. A

Sol. Let OA = r_1, OB = r_2, OC = r_3

$$\frac{1}{r_1^2}+\frac{1}{r_2^2}+\frac{1}{r_3^2}=\frac{1}{4}\left(\cos^2 30^\circ+\cos^2 120^\circ+\cos^2 150^\circ\right)+\left(\sin^2 30^\circ+\sin^2 120^\circ+\sin^2 150^\circ\right)$$

$$=\frac{1}{4}\left(\frac{7}{4}\right)+\frac{5}{4}=\frac{27}{16}$$

72. A

Sol. Equation of focal chord, y = m(x – 2)

$\Rightarrow$ (2m – 6) = $2\sqrt{1+x^2}$ $\Rightarrow 0.m^2 - 6m + 8 = 0,\ m=\frac{4}{3}, \infty$

73. D

Sol. A = {–2, –1, 0, 1}, B = {–3, –2, –1, 0} $\Rightarrow n(A \cap B) = 3$

74. C

Sol. $f(x)=\int_0^x \frac{t^2+1}{t^4+t^2+1}dt$ $\Rightarrow$ f(x) is increasing in $x \in [0, 1] \Rightarrow 0 \le f(x) \le 1$

75. D

Sol. $PP^T = I \Rightarrow \cos^2 A=\frac{1}{2},\ \sin^2 B=\frac{1}{6},\ \tan^2 C=\frac{1}{3}$

76. D

Sol. Circum radius of orthic triangle = $\frac{R}{2}$ and $R=\frac{abc}{4\Delta}=\frac{35}{4\sqrt{6}}$

77. B

Sol. The possible triplets of a, b, c are (1, 4, 8), (1, 3, 11) and (1, 2, 17)

$V_{max} = 34$, $V_{min} = 32$

78. A

Sol. $P_1 = 3\sqrt{3}$, $P_2 = 9$

79. A

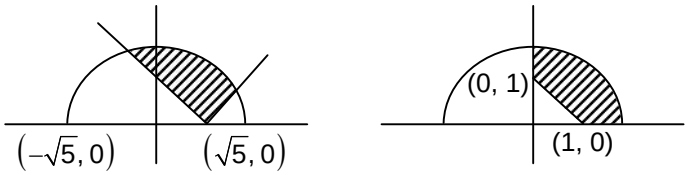
Sol. Let height of hill ℓ

$$\tan\beta=\frac{\ell}{y},\ \tan\alpha=\frac{\ell-10}{y}$$

$$\Rightarrow\ \ell=\frac{10\cos\alpha\sin\beta}{\sin(\beta-\alpha)}$$

80. C

Sol. So, area = $\frac{\pi\left(\sqrt{5}\right)^2}{4}-\frac{1}{2}\times|x|=\frac{5\pi}{4}-\frac{1}{2}$



SECTION – B

81. 00020.00

Sol. Locus will be $\frac{16}{x^2}-\frac{9}{y^2}=4$

82. 00025.00

Sol. ${}^{k}C_r+{}^{k}C_{r-1}={}^{k+1}C_r$

83. 00246.00

Sol. $k_1 + k_{101} = 50,\ \frac{k_{101}-k_1}{k_1k_{101}}=10d \Rightarrow k_1k_{101} = 10$

k is $(k_{101} - k_1)^2 = (k_1 + k_{101})^2 - 4k_1k_{101} = 2460$

84. 00000.60

Sol. Point will be $\left(\frac{1}{5},\frac{-4}{5}\right)$

85. 00001.33

Sol. $p = q = r \Rightarrow R_1=\frac{2}{\sqrt{3}}$

86. 00300.00

Sol. $R(\sin 2A + \sin 2B + \sin 2C) = S \Rightarrow 4R \sin A \sin B \sin C = S$

$\Rightarrow R = 2r$, so triangle is equilateral triangle

87. 00010.24

Sol. $A + B = 2M \Rightarrow |A + B| = 2^5 \Rightarrow tr(A + B) = 10$

88. 00000.12

Sol. $P = 0,\ Q=\frac{1}{8}$

89. 00064.00

Sol. $(x - 1)(x^2 - 2x + a) = 0 ; a < 1$

$\Rightarrow \alpha=1-\sqrt{1-a}$; $\beta = 1$

$\gamma=1+\sqrt{1-a}$

90. 00084.00

Sol. $A=\left\{2+0i,-1+i\sqrt{3},-1-i\sqrt{3}\right\}$

$B=\left\{8+0i,\ 2\sqrt{2}+0i,-2\sqrt{2}+0i\right\}$

So, $-1\pm i\sqrt{3}$ and 8 + 0i are farthest, so $L=\sqrt{84}$